SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1153

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing Techniques (BL4, BL5)
Assessment Tool Weightage: 25%

QUESTIONS: 05 **Total Marks:25**

Annexure A

Error Analysis Task

Code of Conduct:

I D RAGHU VARDHAN__(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D RAGHU VARDHAN

RUBRICS FOR CALCULATION:

Error Analysis Task

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Error Identification	20%	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	30%	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	25%	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	15%	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	10%	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Error Analysis Task

Q1. Scenario: Online Examination System
Modules:
Login, Exam Interface, Result Processing

Question:
Write pseudocode for a **quality assurance testing process** that validates student login and exam access functionality. Include steps for input validation, authentication, error handling, and logging of test results.

Q1. Pseudocode for Quality Assurance Testing Process – Online Examination System
Introduction

An **Online Examination System** consists of three major modules: **Login Module, Exam Interface Module, and Result Processing Module**. Quality Assurance (QA) testing is performed to ensure that only authorized students can log in and access examinations. The testing process must validate user inputs, authenticate credentials, handle errors properly, and record all test results for future analysis. The following pseudocode demonstrates a structured QA testing process for validating student login and exam access functionality.

Pseudocode

```
BEGIN QA_Testing_Process

  DISPLAY "======================================"
  DISPLAY "ONLINE EXAMINATION SYSTEM QA TESTING"
  DISPLAY "======================================"

  SET Total_TestCases = 0
  SET Passed_TestCases = 0
  SET Failed_TestCases = 0

  WRITE_LOG("QA Testing Started")

  -----------------------------------------------
  TEST CASE 1 : INPUT VALIDATION
  -----------------------------------------------

  Total_TestCases = Total_TestCases + 1

  INPUT Student_ID
  INPUT Password

  IF Student_ID IS NULL OR Student_ID IS EMPTY THEN

    DISPLAY "ERROR : Student ID Cannot Be Empty"
    WRITE_LOG("FAIL : Empty Student ID")

    Failed_TestCases = Failed_TestCases + 1

  ELSE IF Password IS NULL OR Password IS EMPTY THEN

    DISPLAY "ERROR : Password Cannot Be Empty"
    WRITE_LOG("FAIL : Empty Password")

    Failed_TestCases = Failed_TestCases + 1

  ELSE

    DISPLAY "Input Validation Successful"
    WRITE_LOG("PASS : Input Validation")

    Passed_TestCases = Passed_TestCases + 1

  END IF
```

```
----------------------------------------------
TEST CASE 2 : LOGIN AUTHENTICATION
----------------------------------------------

Total_TestCases = Total_TestCases + 1

IF Student_ID EXISTS IN DATABASE THEN

  IF Password MATCHES STORED_PASSWORD THEN

    DISPLAY "Authentication Successful"
    WRITE_LOG("PASS : Login Successful")

    Login_Status = TRUE

    Passed_TestCases = Passed_TestCases + 1

  ELSE

    DISPLAY "Invalid Password"
    WRITE_LOG("FAIL : Incorrect Password")

    Login_Status = FALSE

    Failed_TestCases = Failed_TestCases + 1

  END IF

ELSE

  DISPLAY "Student Not Found"
  WRITE_LOG("FAIL : Invalid Student ID")

  Login_Status = FALSE

  Failed_TestCases = Failed_TestCases + 1

END IF


----------------------------------------------
TEST CASE 3 : EXAM ACCESS VALIDATION
----------------------------------------------

Total_TestCases = Total_TestCases + 1

IF Login_Status = TRUE THEN

  IF Student_Registered_For_Exam = TRUE THEN

    DISPLAY "Exam Access Granted"

    WRITE_LOG("PASS : Student Allowed To Enter Exam")

    Exam_Access = TRUE
```

```
    Passed_TestCases = Passed_TestCases + 1
  ELSE
    DISPLAY "Student Not Registered For Exam"
    WRITE_LOG("FAIL : Registration Missing")
    Exam_Access = FALSE
    Failed_TestCases = Failed_TestCases + 1
  END IF
ELSE
  DISPLAY "Login Required Before Exam Access"
  WRITE_LOG("FAIL : Unauthorized Access Attempt")
  Exam_Access = FALSE
  Failed_TestCases = Failed_TestCases + 1
END IF

-----------------------------------------------
TEST CASE 4 : SESSION VALIDATION
-----------------------------------------------

Total_TestCases = Total_TestCases + 1
IF Session_Timeout = TRUE THEN
  DISPLAY "Session Expired"
  WRITE_LOG("FAIL : Session Timeout")
  Failed_TestCases = Failed_TestCases + 1
ELSE
  DISPLAY "Session Active"
  WRITE_LOG("PASS : Session Valid")
  Passed_TestCases = Passed_TestCases + 1
END IF

-----------------------------------------------
TEST CASE 5 : SERVER AVAILABILITY CHECK
```

```
----------------------------------------------
Total_TestCases = Total_TestCases + 1
IF Server_Status = ONLINE THEN
   DISPLAY "Server Available"
   WRITE_LOG("PASS : Server Running")
   Passed_TestCases = Passed_TestCases + 1
ELSE
   DISPLAY "Server Not Available"
   WRITE_LOG("FAIL : Server Down")
   Failed_TestCases = Failed_TestCases + 1
END IF

----------------------------------------------
TEST CASE 6 : DATABASE CONNECTION CHECK
----------------------------------------------
Total_TestCases = Total_TestCases + 1
IF Database_Connection = SUCCESS THEN
   DISPLAY "Database Connected"
   WRITE_LOG("PASS : Database Connected")
   Passed_TestCases = Passed_TestCases + 1
ELSE
   DISPLAY "Database Connection Failed"
   WRITE_LOG("FAIL : Database Error")
   Failed_TestCases = Failed_TestCases + 1
END IF

----------------------------------------------
TEST CASE 7 : RESULT PROCESSING VALIDATION
----------------------------------------------
Total_TestCases = Total_TestCases + 1
IF Exam_Submitted = TRUE THEN
```

```
    IF Marks_Calculated_Correctly = TRUE THEN
      DISPLAY "Result Generated Successfully"
      WRITE_LOG("PASS : Result Processing Successful")
      Passed_TestCases = Passed_TestCases + 1
    ELSE
      DISPLAY "Error In Result Calculation"
      WRITE_LOG("FAIL : Incorrect Marks Calculation")
      Failed_TestCases = Failed_TestCases + 1
    END IF
  ELSE
    DISPLAY "Exam Not Submitted"
    WRITE_LOG("FAIL : Missing Submission")
    Failed_TestCases = Failed_TestCases + 1
  END IF

  ------------------------------------------------ ERROR HANDLING SECTION ------------------------------------------------

  TRY
    CONNECT DATABASE
  CATCH Database_Exception
    DISPLAY "Database Exception Occurred"
    WRITE_LOG("ERROR : Database Failure")
  END TRY

  TRY
    CONNECT SERVER
  CATCH Server_Exception
    DISPLAY "Server Exception Occurred"
```

```
    WRITE_LOG("ERROR : Server Failure")

  END TRY


  -----------------------------------------------
  TEST SUMMARY REPORT
  -----------------------------------------------

  DISPLAY "=================================="
  DISPLAY "QA TESTING SUMMARY REPORT"
  DISPLAY "=================================="

  DISPLAY "Total Test Cases : ", Total_TestCases
  DISPLAY "Passed Test Cases : ", Passed_TestCases
  DISPLAY "Failed Test Cases : ", Failed_TestCases

  WRITE_LOG("Total Tests = " + Total_TestCases)
  WRITE_LOG("Passed Tests = " + Passed_TestCases)
  WRITE_LOG("Failed Tests = " + Failed_TestCases)

  IF Failed_TestCases = 0 THEN

    DISPLAY "SYSTEM READY FOR DEPLOYMENT"

    WRITE_LOG("FINAL RESULT : PASS")

  ELSE

    DISPLAY "SYSTEM REQUIRES BUG FIXING"

    WRITE_LOG("FINAL RESULT : FAIL")

  END IF

  WRITE_LOG("QA Testing Completed")

END QA_Testing_Process
```

Explanation of the Testing Process

1. Input Validation

The system verifies whether the Student ID and Password fields are empty or invalid. Invalid inputs are rejected immediately and recorded in the test log.

2. Authentication Testing

The entered credentials are compared with the database records. Successful authentication allows access, while invalid credentials generate appropriate error messages.

3. Exam Access Validation

After successful login, the system checks whether the student is registered for the examination. Only registered students can access the exam interface.

4. Session Validation

The system checks whether the user's session is active or expired. Expired sessions prevent unauthorized access.

5. Server Availability Testing

QA verifies that the examination server is online and capable of handling requests.

6. Database Connection Testing

The database connection is tested to ensure login verification and result processing can occur without interruption.

7. Result Processing Validation

After exam submission, the system verifies that marks are calculated correctly and results are generated accurately.

8. Error Handling

Exception-handling mechanisms are tested to ensure the system responds gracefully to server or database failures.

9. Logging of Test Results

Every test case result (PASS, FAIL, ERROR) is recorded in a log file for auditing, debugging, and quality analysis purposes.

Flow Diagram

```
        START
          |
          v
    Enter Login Details
          |
          v
    Input Validation
          |
     ----------------
     |          |
   Invalid     Valid
     |          |
     v          v
 Log Failure  Authentication
                |
          ----------------
          |          |
        Invalid     Valid
          |          |
          v          v
      Log Failure   Check Exam
                  Registration
                     |
               ----------------
               |          |
            Not Reg.    Registered
               |          |
               v          v
          Log Failure   Grant Access
                          |
                          v
                    Process Exam
                          |
                          v
                   Generate Result
```

```
        |
        v
  Log Test Results
        |
        v
       END
```

Conclusion

The above pseudocode provides a comprehensive Quality Assurance testing process for an Online Examination System. It validates student login credentials, checks exam access permissions, performs input validation, handles errors effectively, verifies result processing, and logs all testing outcomes. This systematic approach ensures reliability, security, accuracy, and smooth operation of the examination platform